

OIC Stock Alert - Low-Level Design

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Source Design: docs/02-design/features/oic-stock-alert.design.md
Repository: <https://github.com/ai-ml-kiosk/test-oic-stock-api>

1. Purpose

This Low-Level Design defines the detailed contracts and implementation logic for the OIC Stock Alert API. It covers JSON payload specifications, mapping logic, validation rules, OIC Switch branch structures, module-level behavior, and test cases.

The companion HLD is:

docs/02-design/features/oic-stock-alert.hld.md

2. Source Artifacts

Artifact | Purpose
artifacts/openapi/oic-stock-alert.openapi.json | REST API contract.
artifacts/json-schema/alert-evaluation-request.schema.json | Request payload schema.
artifacts/json-schema/alert-evaluation-response.schema.json | Success and error response schema.
artifacts/oic/oic-switch-branches.md | OIC Switch expression and branch mapping guide.
artifacts/samples/*.json | Example request, success, no-alert, validation error, and timeout payloads.

3. Endpoint Detail

3.1 'GET /health'

Response status:

- 200 OK

Response fields:

Field	Type	Description
status	string	ok when service is running.
service	string	oic-stock-alert.
version	string	API version.
providerMode	string	Active quote provider mode.
timestamp	string	UTC timestamp in ISO-8601 format.

3.2 'POST /v1/alerts/evaluate'

Request headers:

Header	Required	Description
Content-Type	Yes	application/json JSON payload format.
X-Request-Id	No	Caller-provided correlation ID fallback.

Status mapping:

Condition	HTTP Status
Valid evaluation, alert triggered	200
Valid evaluation, no alert	200
Invalid JSON or validation failure	400

Unsupported rule configuration | 422
 Quote not found or provider error | 502
 Provider timeout | 504
 Unexpected exception | 500

4. Request JSON Payload

Canonical request:

```
{
  "requestId": "oic-run-20260525-001",
  "source": {
    "system": "OIC",
    "integrationName": "StockAlertIntegration",
    "instanceId": "300000123456789"
  },
  "symbol": "ORCL",
  "currency": "USD",
  "rules": [
    {
      "ruleId": "orcl-above-150",
      "metric": "lastPrice",
      "operator": "gte",
      "threshold": 150.0,
      "severity": "high",
      "message": "ORCL crossed target price"
    }
  ],
  "options": {
    "providerMode": "mock",
    "includeDiagnostics": true
  }
}
```

Field rules:

Field	Required	Type	Rule
requestId	No	string	Max 80 characters. Body value wins over header fallback.
source.system	No	string	Expected OIC for OIC calls.
source.integrationName	No	string	Max 120 characters.
source.instanceId	No	string	Copied to oic.trackingId when present.
symbol	Yes	string	Trimmed, uppercased, pattern <code>^[A-Z][A-Z0-9.-]{0,14}\$</code> .
currency	No	string	3-letter code, uppercased.
rules	Yes	array	Minimum 1, maximum 10.
rules[].ruleId	Yes	string	Unique per request, max 80 characters.
rules[].metric	Yes	string	lastPrice, changePercent, dayHigh, dayLow, or volume.
rules[].operator	Yes	string	gt, gte, lt, lte, eq, or absGte.
rules[].threshold	Yes	number	Finite numeric value. Non-negative except for changePercent.
rules[].severity	No	string	info, warning, high, or critical; defaults to warning.
rules[].message	No	string	Max 240 characters; defaults to empty string.
options.providerMode	No	string	mock or live. Runtime config controls override support.
options.includeDiagnostics	No	boolean	Defaults to false.

4.1 Local '.env' Configuration Payload

Local live-provider testing uses an untracked .env file. The committed artifact is .env.example with placeholder values only.

```
QUOTE_PROVIDER_MODE=live
```

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```
QUOTE_PROVIDER_NAME=alpha_vantage
QUOTE_PROVIDER_API_KEY=replace-with-local-alpha-vantage-key
QUOTE_PROVIDER_TIMEOUT_MS=1500
ALLOW_REQUEST_PROVIDER_OVERRIDE=false
```

Parser rules:

- Accept KEY=value lines.
- Trim surrounding whitespace around keys and values.
- Ignore blank lines and lines beginning with #.
- Do not execute shell commands, substitutions, or nested variable expansion.
- Process environment variables override .env values.
- Do not expose QUOTE_PROVIDER_API_KEY in diagnostics, logs, exceptions, or responses.

5. Response JSON Payloads

5.1 Success Response

```
{
  "requestId": "oic-run-20260525-001",
  "status": "success",
  "symbol": "ORCL",
  "quote": {
    "symbol": "ORCL",
    "lastPrice": 152.31,
    "currency": "USD",
    "change": 2.14,
    "changePercent": 1.43,
    "dayHigh": 153.2,
    "dayLow": 149.7,
    "volume": 12654321,
    "asOf": "2026-05-25T05:30:00Z",
    "provider": "mock"
  },
  "rules": [
    {
      "ruleId": "orcl-above-150",
      "metric": "lastPrice",
      "operator": "gte",
      "threshold": 150.0,
      "actualValue": 152.31,
      "triggered": true,
      "severity": "high",
      "message": "ORCL crossed target price"
    }
  ],
  "decision": {
    "triggered": true,
    "code": "ALERT_TRIGGERED",
    "severity": "high",
    "message": "1 alert rule triggered for ORCL"
  },
  "oic": {
    "switchBranch": "ALERT_TRIGGERED",
    "notificationRecommended": true,
    "retryRecommended": false,
    "trackingId": "300000123456789"
  },
}
```

```

    "diagnostics": {
      "providerMode": "mock",
      "elapsedMs": 48,
      "ruleCount": 1,
      "triggeredCount": 1
    }
  }
}

```

5.2 Error Response

```

{
  "requestId": "oic-run-20260525-003",
  "status": "error",
  "symbol": "BAD SYMBOL",
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Request validation failed",
    "details": [
      {
        "field": "symbol",
        "issue": "Symbol must match pattern ^[A-Z][A-Z0-9.-]{0,14}$"
      }
    ]
  },
  "decision": {
    "triggered": false,
    "code": "INPUT_ERROR",
    "severity": "warning",
    "message": "Input could not be evaluated"
  },
  "oic": {
    "switchBranch": "INPUT_ERROR",
    "notificationRecommended": false,
    "retryRecommended": false,
    "trackingId": "300000123456791"
  }
}

```

6. Mapping Logic

6.1 Request Normalization

Source | Target | Logic

Body requestId | request.requestId | Use when supplied and valid.

Header X-Request-Id | request.requestId | Fallback when body value is absent.

Generated UUID | request.requestId | Fallback when body and header are absent.

Body symbol | request.symbol | Trim, uppercase, validate pattern.

Body currency | request.currency | Uppercase; if absent, use provider quote currency or USD.

source.instanceId | oic.trackingId | Copy when present; otherwise use requestId.

options.providerMode | provider mode | Apply only when request override is enabled; otherwise use runtime default.

Missing rules[].severity | normalized rule severity | Default to warning.

Missing rules[].message | normalized rule message | Default to empty string.

6.1.1 Alpha Vantage Outbound Request Mapping

Provider selection:

Setting | Required Value

QUOTE_PROVIDER_MODE | live

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QUOTE_PROVIDER_NAME | alpha_vantage
QUOTE_PROVIDER_API_KEY | Non-empty local or runtime secret

Outbound request:

GET https://www.alphavantage.co/query

Query parameters:

Parameter	Value
function	GLOBAL_QUOTE
symbol	Normalized request symbol
apikey	QUOTE_PROVIDER_API_KEY
datatype	Omitted; JSON is the default
entitlement	Omitted for MVP

Timeout:

- Use QUOTE_PROVIDER_TIMEOUT_MS.
- Default: 1500.
- Timeout maps to PROVIDER_TIMEOUT.

Freshness:

- Alpha Vantage quote freshness depends on API entitlement.
- Realtime or 15-minute delayed data is outside the default MVP guarantee.

6.1.2 Alpha Vantage Response Normalization

Alpha Vantage Field	Internal Field	Conversion
01. symbol	symbol	String.
03. high	dayHigh	Decimal string to number.
04. low	dayLow	Decimal string to number.
05. price	lastPrice	Decimal string to number.
06. volume	volume	Integer string to integer.
07. latest trading day	asOf	Date string to UTC date-like timestamp.
09. change	change	Decimal string to number.
10. change percent	changePercent	Strip %, then decimal string to number.
constant	currency	USD for MVP.
constant	provider	alpha_vantage.

6.2 Rule Evaluation

Operator	Trigger Logic
gt	actualValue > threshold
gte	actualValue >= threshold
lt	actualValue < threshold
lte	actualValue <= threshold
eq	actualValue == threshold after numeric normalization
absGte	abs(actualValue) >= threshold

The evaluator reads the requested metric from the quote. If the metric is missing from a quote that should contain it, the provider path is mapped to a quote-unavailable style error.

6.3 Aggregate Decision

Condition	Decision Code	Triggered	Severity	Switch Branch	Retry
One or more rules triggered	ALERT_TRIGGERED	true	Highest triggered severity		
ALERT_TRIGGERED		false			
No rules triggered	NO_ALERT	false	info	NO_ALERT	false
Validation failure	INPUT_ERROR	false	warning	INPUT_ERROR	false
Unsupported rule	INPUT_ERROR	false	warning	INPUT_ERROR	false

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```
Quote not found | QUOTE_UNAVAILABLE | false | warning | QUOTE_UNAVAILABLE | false
Provider timeout | PROVIDER_TIMEOUT | false | warning | PROVIDER_TIMEOUT | true
Unexpected exception | SYSTEM_ERROR | false | critical | SYSTEM_ERROR | true
Alpha Vantage empty quote | QUOTE_UNAVAILABLE | false | warning | QUOTE_UNAVAILABLE | false
Alpha Vantage rate-limit note | QUOTE_UNAVAILABLE | false | warning | QUOTE_UNAVAILABLE | OIC policy
Alpha Vantage invalid-key information | QUOTE_UNAVAILABLE | false | warning | QUOTE_UNAVAILABLE | false
```

Severity precedence:

critical > high > warning > info

7. OIC Switch Branch Structures

Primary expression:

```
response.oic.switchBranch
```

Optional assigned variable:

```
stockAlertSwitchBranch = response.oic.switchBranch
```

Branches:

Branch	Match Value	Retry	Notification	Purpose
Alert Triggered	ALERT_TRIGGERED	No	Yes	Build and send alert notification.
No Alert	NO_ALERT	No	No	Log no-op and end successfully.
Input Error	INPUT_ERROR	No	No	Record invalid inbound payload or unsupported rule.
Quote Unavailable	QUOTE_UNAVAILABLE	Usually no	No	Record business exception for missing market data.
Provider Timeout	PROVIDER_TIMEOUT	Yes	No	Route to retry or recoverable fault handling.
System Error	SYSTEM_ERROR	Yes	No	Route to technical fault handling and operations logging.
Otherwise	unmatched	No	No	Treat as technical fault and preserve diagnostics.

Alert notification mapping:

Target Field	Source
subject	concat(response.symbol, " stock alert: ", response.decision.severity)
body	response.decision.message plus triggered rule messages and quote values
symbol	response.symbol
price	response.quote.lastPrice
currency	response.quote.currency
triggeredRules	Filter response.rules where triggered = true
trackingId	response.oic.trackingId
requestId	response.requestId

Error logging mapping:

Branch	Primary Fields
INPUT_ERROR	requestId, symbol, error.code, error.details
QUOTE_UNAVAILABLE	requestId, symbol, error.code, error.message
PROVIDER_TIMEOUT	requestId, symbol, error.code, oic.retryRecommended
SYSTEM_ERROR	requestId, symbol, error.code, error.message

8. Module-Level Design

Module	Key Functions or Types	Behavior
--------	------------------------	----------

config.py | Settings, settings loader | Reads environment defaults for port, provider, timeout, and override behavior.
validation.py | request validation and normalization | Enforces JSON shape, symbols, rule count, unique IDs, metrics, operators, thresholds, severity, and options.
quote_provider.py | mock quote provider | Returns deterministic quote data and provider error fixtures for contract tests.
alpha_vantage_provider.py | Alpha Vantage REST client | Calls GLOBAL_QUOTE, applies timeout, normalizes response, and maps provider errors.
env_loader.py | local .env parser | Loads untracked local settings without shell execution or secret logging.
evaluator.py | evaluate_rules, build_decision | Applies rule operators and derives aggregate decision code, severity, and message.
mapper.py | build_success_response, build_error_response | Builds OIC-friendly response payloads with decision and branch fields.
errors.py | stock alert exception classes | Converts validation, provider, timeout, and system conditions into stable error metadata.
app.py | application service | Coordinates validation, provider lookup, evaluation, mapping, and error handling.
server.py | HTTP handler | Parses requests, routes endpoints, serializes JSON responses, and emits status codes.

9. Error Contract

Error Code	HTTP Status	Decision Code	Switch Branch	Retry
VALIDATION_ERROR	400	INPUT_ERROR	INPUT_ERROR	No
UNSUPPORTED_RULE	422	INPUT_ERROR	INPUT_ERROR	No
QUOTE_NOT_FOUND	502	QUOTE_UNAVAILABLE	QUOTE_UNAVAILABLE	No
PROVIDER_ERROR	502	QUOTE_UNAVAILABLE	QUOTE_UNAVAILABLE	Maybe by OIC policy
PROVIDER_TIMEOUT	504	PROVIDER_TIMEOUT	PROVIDER_TIMEOUT	Yes
SYSTEM_ERROR	500	SYSTEM_ERROR	SYSTEM_ERROR	Yes

Alpha Vantage provider mapping:

Upstream Payload or Condition	Error Code	Switch Branch	Notes
Empty Global Quote object	QUOTE_NOT_FOUND	QUOTE_UNAVAILABLE	Covers invalid or unsupported symbol payloads.
Note message	PROVIDER_ERROR	QUOTE_UNAVAILABLE	Covers provider rate-limit messages.
Information message	PROVIDER_ERROR	QUOTE_UNAVAILABLE	Covers invalid key or entitlement-style messages.
Outbound timeout	PROVIDER_TIMEOUT	PROVIDER_TIMEOUT	Must set oic.retryRecommended=true.
Malformed JSON	PROVIDER_ERROR	QUOTE_UNAVAILABLE	Do not expose raw provider payload if it might include secrets.
Unexpected client exception	SYSTEM_ERROR	SYSTEM_ERROR	Log sanitized exception category only.

10. Validation Rules

- Reject malformed JSON.
- Reject oversized JSON bodies beyond the service limit.
- Require symbol.
- Require one to ten rules.
- Reject duplicate ruleId values.
- Reject unknown metrics.
- Reject unknown operators.
- Reject missing or non-numeric thresholds.
- Reject negative thresholds except where supported for changePercent.
- Validate severity against info, warning, high, and critical.

- Default missing severity to warning.
- Exclude diagnostics unless `options.includeDiagnostics` is true.
- Reject live mode when `QUOTE_PROVIDER_API_KEY` is missing or blank.
- Reject unsupported live provider names.
- Never include `QUOTE_PROVIDER_API_KEY` in error details.

11. Test Cases

11.1 Unit Tests

Test | Expected Result

lastPrice gte threshold with actual above threshold | Rule triggered.
lastPrice lt threshold with actual above threshold | Rule not triggered.
changePercent absGte threshold with negative actual beyond threshold | Rule triggered.
Multiple triggered rules with different severities | Aggregate severity is highest severity.
No triggered rules | `decision.code = NO_ALERT`.
Duplicate ruleId values | Validation error.
Unsupported metric | Input error path.
.env parser ignores comments and blank lines | Settings load successfully.
Process environment overrides .env | Effective runtime setting comes from process environment.
Missing Alpha Vantage API key in live mode | Provider error without secret leakage.
Alpha Vantage Global Quote fixture | Normalized quote has numeric lastPrice, dayHigh, dayLow, change, changePercent, and integer volume.
Alpha Vantage rate-limit Note fixture | `QUOTE_UNAVAILABLE` branch.
Alpha Vantage invalid-key Information fixture | `QUOTE_UNAVAILABLE` branch.
Alpha Vantage timeout | `PROVIDER_TIMEOUT` branch and retry recommended.
Secret redaction | API key absent from logs, diagnostics, response body, and assertion output.

11.2 API Contract Tests

Scenario | Expected Result

GET /health | 200 with service metadata.
Valid alert-triggered request | 200, `decision.code = ALERT_TRIGGERED`, `oic.switchBranch = ALERT_TRIGGERED`.
Valid no-alert request | 200, `decision.code = NO_ALERT`, `oic.switchBranch = NO_ALERT`.
Invalid symbol | 400, `error.code = VALIDATION_ERROR`, `oic.switchBranch = INPUT_ERROR`.
Mock provider quote missing | 502, `oic.switchBranch = QUOTE_UNAVAILABLE`.
Simulated provider timeout | 504, `oic.switchBranch = PROVIDER_TIMEOUT`, `retryRecommended = true`.
Live Alpha Vantage fixture alert-triggered | 200, provider `alpha_vantage`, expected branch from evaluated quote.
Live Alpha Vantage invalid symbol fixture | 502, `oic.switchBranch = QUOTE_UNAVAILABLE`.

11.3 Artifact Tests

Artifact | Expected Result

OpenAPI JSON | Valid JSON and includes `/v1/alerts/evaluate`.
Request schema | Valid JSON and requires symbol and rules.
Response schema | Valid JSON and defines success and error response alternatives.
Sample payloads | Valid JSON and aligned with branch expectations.
OIC switch guide | Documents all branch values.

11.4 Verification Command

```
python3 -m unittest discover -s tests
```


12. Implementation Sequence

1. Load runtime settings.
2. Parse HTTP request and JSON body.
3. Normalize and validate request.
4. Load .env and process environment into settings.
5. Select provider mode.
6. If live Alpha Vantage mode is selected, require API key and call GLOBAL_QUOTE.
7. Normalize provider quote into internal shape.
8. Evaluate rules.
9. Build aggregate decision.
10. Map success or error response.
11. Return HTTP status and JSON response.
12. Let OIC branch on response.oic.switchBranch.

13. LLD Decisions

Decision | Rationale

Keep decision.code equal to success oic.switchBranch | Prevents branch mismatch for successful evaluations.

Always include oic routing object | OIC can branch consistently on success and error responses.

Keep diagnostics optional | Reduces payload noise while preserving troubleshooting support.

Preserve requestId in all responses | Enables retry correlation and support investigation.

Use explicit switch branch enum | Avoids brittle text matching in OIC.

Keep Alpha Vantage behind provider interface | Live market data changes do not alter OIC-facing API contracts.

Use .env only as local secret input | Prevents accidental API-key commits while preserving deployed environment-variable configuration.

Omit Alpha Vantage entitlement for MVP | Keeps first live integration small; realtime and delayed entitlement can be added later.